

Demonstrating Jamming Mitigation in O-RAN via AI enabled Intrusion Detection and Secure Slicing xApps

Joshua Moore, Aly Sabri Abdalla, Charles Ueltschey, Vuk Marojevic

Department of Electrical and Computer Engineering, Mississippi State University, MS 39762, USA

Email: {jjm702, asa298, cmu32, vuk.marojevic}@msstate.edu

Abstract—This demo illustrates a coordinated jamming mitigation strategy in Open RAN using three near-real-time RIC xApps: a KPI monitoring xApp, an intrusion detection xApp with a conditional meta-model, and a secure slicing xApp. The KPI xApp collects periodic RAN metrics and forwards them to the detection xApp, which classifies jamming per UE using either a supervised or unsupervised model based on UE distance. Upon detection, the slicing xApp dynamically reallocates physical resource blocks at the gNB to shift affected UEs to cleaner spectrum, preserving service without handovers. The demo highlights the agility and programmability of O-RAN for real-time, intelligent physical-layer defense.

Index Terms—O-RAN, Security, Jamming Mitigation, AI

I. INTRODUCTION

As Open RAN (O-RAN) architectures enable greater openness and programmability, they also expose new security risks, particularly at the radio access layer. A key threat is physical-layer jamming, where adversaries disrupt communication by transmitting interference that corrupts control or data channels, degrades signal quality, or prevents successful decoding at the receiver. Real-time detection and mitigation of such attacks is essential for maintaining secure and reliable wireless service [1].

This demonstration introduces a coordinated O-RAN jamming mitigation strategy using three near-real time (near-RT) radio intelligent controller (RIC) xApps detailed in Fig. 1a. The first xApp is the Kpimoni that collects key performance indication (KPI) reports every 10 ms and stores them in a database. The intrusion detection (ID)-xApp employs a dual model meta approach to detecting jamming via patterns in KPI reports. The third xApp, Secure Slicing (SS-xApp), dynamically reallocates physical resource blocks (PRBs), shifting UEs to safe frequency bands. Together, these xApps form an intelligent, closed-loop control system enabled by the RIC reacting to jamming as opposed to previous demonstrations focusing on offensive mitigations [2].

Experiments were conducted on a 5G setup leveraging srsRAN and two COTS UEs, one streaming YouTube, the other idle. UEs were positioned at 1, 5, 10, 20, 30, and 40 ft from the gNB under both normal and jammed conditions, with the jammer location and gain varied. Metrics collected included UE distance, jammer parameters, and PHY-layer indicators (Physical Cell Identity (PCI), Channel quality indicator (CQI), Rank Indicator (RI), Modulation Cod-

ing Scheme (MCS)), along with throughput (dl_brat, ul_brat), reliability (dl_ok, dl_nok, dl_percent), and control signals (BSR, TA, PHR).

Both models leverage a set of engineered features crucial for accurate detection, including cqi_diff for instantaneous CQI volatility, cqi_rolling_std for sustained CQI instability, and quality_divergence as a physical contradiction metric that contrasts RSRP signal strength in the uplink with CQI as a signal quality metric. This hybrid feature set enables the xApp to capture sudden fluctuations, persistent interference, and differentiate between low signal quality resulting from bad channel and active jamming scenarios.

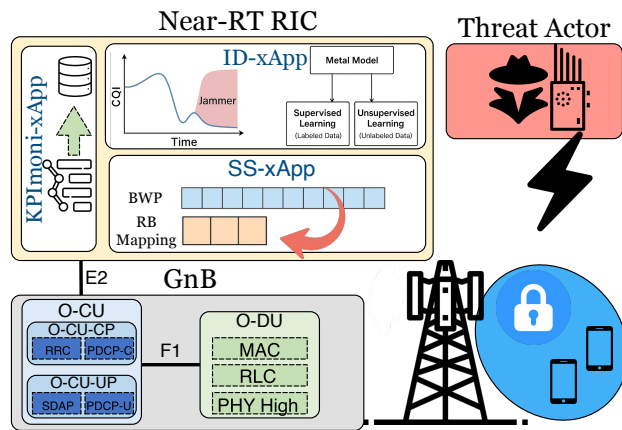
The dataset is streamed as KPM reports to the near-RT RIC, enabling the ID-xApp to infer jamming likelihood per UE. Once a confidence threshold is crossed, it triggers the SS-xApp to reassign PRBs in real time. The demo highlights this closed-loop defense without requiring live radio hardware.

The rest of the paper is organized as follows: Section II describes the ID-xApp and its meta-model approach; Section III explains the SS-xApp; and Section IV concludes with future work.

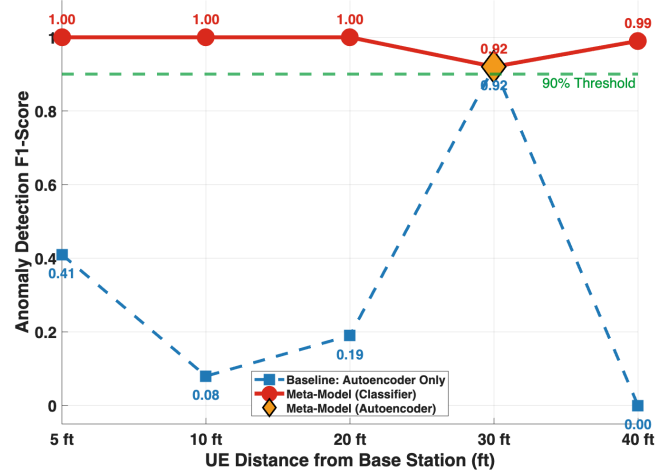
II. RNN-BASED INTRUSION DETECTION

The ID-xApp uses a conditional distance-aware meta-model framework that switches between two LSTM-based sub-models based on UE distance. At mid-range (around 30 ft), where signal degradation is complex due to the varying channel conditions, including fading and path loss complemented with jamming interference, it employs a deep LSTM Autoencoder to detect jamming via reconstruction error. At other distances where the UE is very close or far from the gNB, where the channel is more stable, either as a good channel near the gNB or a bad channel far from the gNB, the model selects a supervised LSTM classifier, which better handles clear signal patterns or severe degradation.

Fig. 1b shows that the autoencoder-only baseline performs inconsistently over UE distances, with F1-scores of 0.41 at 5 ft, 0.08 at 10 ft, 0.19 at 20 ft, peaking at 0.92 at 30 ft, and dropping to 0.00 at 40 ft. In contrast, the meta-model adaptively selects the best model for each range: the classifier yields consistent F1-scores of 1.00 (5–20 ft) and 0.99 (40 ft), while the autoencoder is used only at 30 ft. Classifiers excel



(a) System architecture showing coordinated use of three xApps within the Near-RT RIC



(b) Anomaly detection F1-score of the autoencoder baseline and meta-model classifier across different UE distances from the base station

Fig. 1: Meta-Model-Driven Jamming Detection and Mitigation in Near-Real-Time RIC

when signal features are distinct, either clean (close) or highly impaired (far), making anomalies easier to classify. The autoencoder is most effective at transitional distances, where nuanced temporal patterns reveal jamming effects [3], allowing for robust detection across all scenarios.

III. SECURE FREQUENCY SLICING FOR JAMMING MITIGATION

Upon jamming detection by the ID-xApp, the SS-xApp is triggered to dynamically reallocate PRBs beyond the RBs under attack, isolating the affected frequency bands. The SS-xApp interfaces with the gNB scheduler via the E2 interface to adjust the UE downlink allocations in real time, redirecting traffic to more secure and reliable spectrum slices without service disruption.

The xApp leverages Bandwidth Parts (BWPs), using the configurable parameters `start_rb` and `max_rb_size` to define contiguous PRB ranges within the active BWP. It performs a sliding-window search across candidate PRB sets, excluding those flagged as jammed based on intrusion classification and recent CQI. When available, signal quality metrics guide selection of the least-interfered parts; otherwise, the window farthest from interference is chosen. The process supports sub-second response times, preserving link reliability and near-real-time mitigation of jamming attacks without disrupting active service continuity.

IV. CONCLUSION AND FUTURE WORK

This demonstration highlights the effectiveness of leveraging AI-enabled intrusion detection combined with secure frequency slicing xApps for real-time jamming mitigation in O-RAN. By analyzing KPI reports with a meta-model, the system accurately identifies jamming attacks and promptly reallocates PRBs to protect legitimate UEs for various ranges

of distance. The approach showcases the power of O-RAN's programmable architecture and near-RT RIC platform in enabling intelligent, automated responses to physical-layer threats, paving the way for more resilient and secure wireless networks. Future work will include a full analysis of the conditional meta-model and the development of a dynamic sub-distance confidence threshold to further optimize detection accuracy and response decisions.

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